

THANUSH RAJ UMA BABU

Langfeldstraße 1, 91058 Erlangen, Germany · thanushraj23@gmail.com · +49 170 8807756 · [linkedin.com/in/thanush-raj](https://www.linkedin.com/in/thanush-raj)

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Dear Hiring Manager,

I am writing to express my interest in a Werkstudent position in Data Science / Machine Learning at your company. I am currently pursuing my M.Sc. in Data Science with a specialisation in Artificial Intelligence and Machine Learning at Friedrich-Alexander-Universität Erlangen-Nürnberg, and I bring a combination that is still uncommon among working students: published academic research in machine learning and two and a half years of enterprise engineering experience.

For my Bachelor's thesis, I designed and published a content-based recommendation system (NCECA 2023, DOI: 10.5281/zenodo.7965018) built on dual neural networks with a custom cost function and L2 regularisation, evaluated against collaborative and hybrid baselines using AUC, log loss, and recall. I also implemented a retrieval-and-ranking pipeline to keep inference efficient across large item catalogues — the kind of practical engineering constraint that matters as much in production as in research.

Before returning to academia, I worked as a Technical Engineer at Kyndryl, where I managed and monitored ETL batch workflows across four enterprise clients (Honda, TATA Steel, ASR, and Ethniki Insurance) while maintaining 99.9% pipeline reliability. I led the end-to-end transition of Ethniki Insurance to Kyndryl's digital workplace and mentored four junior engineers to full operational independence. That role taught me to reason carefully about data integrity, to write documentation other people can actually follow, and to take ownership of unfamiliar problems.

My toolkit includes Python (NumPy, Pandas, PyTorch), Transformers (Hugging Face), SQL, Tableau, Docker, and Git/GitHub for collaborative, version-controlled development. I hold certifications including Microsoft Azure AI Fundamentals (AI-900), the Google Data Analytics Professional Certificate, and Stanford's Supervised Machine Learning course. I am based in Erlangen, speak English at C1 level, and am actively improving my German (currently A2).

As a Werkstudent, I can commit up to 20 hours per week during the semester and full-time during semester breaks. I would welcome the opportunity to contribute to your team and to discuss how my background in machine learning and data engineering could support your work.

Thank you for your time and consideration.

Warm regards,

Thanush Raj Uma Babu